What changes an asteroid’s path?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how gravity can change velocity without contact.

SUCCESS CRITERIA - I can...

1. Draw velocity tangent to a path and force toward the attracting body.
2. Distinguish motion from a change in motion.
3. State what a prediction of straight-line motion leaves out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “acceleration” to explain your answer. Sentence starters are optional.

gravity: an attractive interaction between masses

acceleration: rate of change of velocity

model: a drawing, explanation or equation for a physical situation

01 Warm-up: bring back an earlier idea

A simplified calculation predicts position (**-30**, **50**) km with bounds ±**3 km** in x and ±**5 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

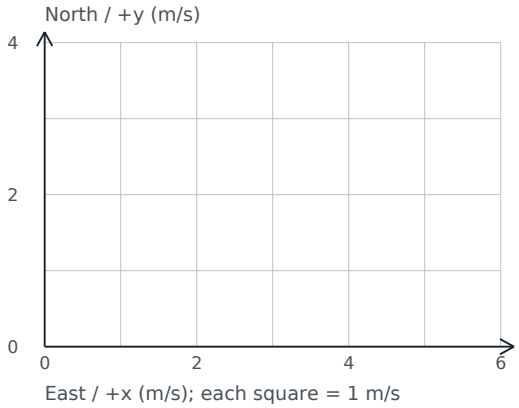
02 Worked example: explain the prediction

Reference: $a = F_{net}/m$.
Gravity attracts masses and can change speed, direction or both. Our asteroid scenario is fictional.

Worked reasoning: A fictional asteroid has a velocity arrow pointing right. A brief upward force gives it an upward change in velocity. The new velocity combines the original rightward motion with this change.

Your turn: A spacecraft is initially moving right. A short thruster burn adds an upward change in velocity.
1. **Draw the original velocity** and the change head-to-tail.
2. **Draw the final velocity** from the starting tail to the ending tip.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

What changes an asteroid’s path?

antocciphysics

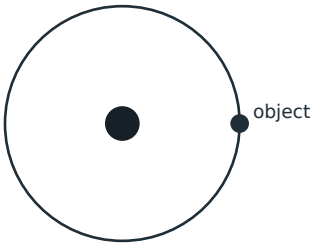
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

1. **Sketch the velocity** and gravitational-force arrows at the marked point on the orbit.
2. Why do they point in different directions?

Sentence starter: Both … . They differ in … because … .



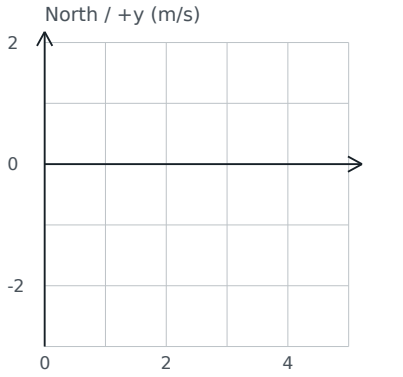
The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

A simulated probe starts with velocity (**4**, **0**) m/s. Its thruster produces acceleration (**0**, **-1**) m/s² for **2 s**. Plot the initial and final velocity vectors. **Find the final speed** and **explain whether** speed stayed constant.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says space has no gravity because astronauts float. **Explain how orbiting astronauts can** be falling.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

What changes an asteroid’s path?

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A **0.5 kg** system accelerates at **0.7 m/s²**. **Predict acceleration** for **0.9 kg** at the same net force. **Explain the effect of greater** resistance.
Use the table to keep the controlled and changed quantities distinct.

Word help: system = the object or objects chosen for analysis

Sentence starter: Both … . They differ in … because … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

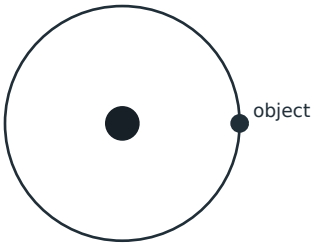
Marzano check - circle: ① ② ③

07 Exit check

A force stays perpendicular to velocity during ideal circular motion. **Explain what changes** and what stays constant.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

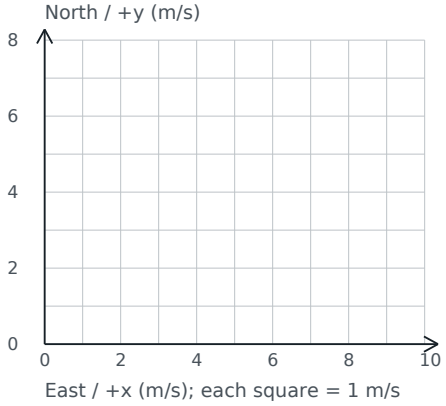


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A simulated probe starts with velocity (**8**, **0**) m/s. Its thruster produces acceleration (**0**, **2**) m/s² for **3 s**.
1. **Draw the change in velocity** head-to-tail with the initial velocity.
2. **Find the final components** and speed.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③